

DSAA3073: Theories in Data Science

Week 2: PAC Learning & VC Dimension

Concise Learning Sheet

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Explorative projection

Introduction: Why Can't We Just Minimize Training Error?

Machine learning seems simple: collect data, find a hypothesis that fits the data well, and deploy it. But this intuitive approach hides subtle dangers that can lead to catastrophic failures in practice.

This week, we develop the mathematical foundations to answer:

1. **Why isn't training accuracy enough?** (PAC framework)
2. **Why must convergence be uniform?** (Selection bias)
3. **How do we measure complexity?** (VC dimension)
4. **Why are infinite classes learnable?** (Sauer's lemma)

By the end, you'll understand the complete theory connecting geometry (VC dimension), combinatorics (growth functions), and learning (sample complexity).

This Week's Learning Objectives

After completing this material, you should be able to:

1. **PAC Learning Framework:**
 - Explain why training error alone is insufficient
 - Define PAC learnability with precise guarantees
 - Compute sample complexity for simple settings
2. **Uniform Convergence:**
 - Identify when pointwise convergence fails
 - Apply union bounds correctly
 - Distinguish selection bias from valid inference
3. **VC Dimension:**
 - Compute VC dimension for hypothesis classes
 - Construct shattering configurations (lower bounds)
 - Prove impossibility arguments (upper bounds)
4. **Sauer's Lemma:**
 - Explain the exponential-to-polynomial transformation
 - Apply VC bounds to sample complexity
 - Connect geometry to learning guarantees

The PAC Learning Framework: Why Training Accuracy Isn't Enough

🧐 Student Question

From the class: “I trained my neural network on 1000 images and got 99% training accuracy. That’s amazing! Can I deploy this model in production?”

This seems like success—if the model performs well on training data, shouldn’t it perform well on new data too?

The Natural Approach: Minimize Training Error

The most intuitive learning strategy is straightforward: find the hypothesis that makes the fewest mistakes on your training data.

Strategy: Given training set $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$, choose:

$$\hat{h} = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h) = \operatorname{argmin}_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \mathbb{1}[h(x_i) \neq y_i] \quad (1)$$

This is called **Empirical Risk Minimization (ERM)**—minimize the empirical (observed) error.

Intuition: If $\hat{h}$ performs well on the sample, it should perform well on new data drawn from the same distribution. After all, the training data is representative, right?

Testing the Intuition: A Concrete Example

Let’s test this with a real scenario.

Setup:

- Dataset: 100 points in $\mathbb{R}^2$ for binary classification
- True pattern: Points inside a circle are positive, outside are negative
- Training sample: 100 randomly drawn points

Model A: Simple (Linear Classifier)

- Hypothesis class: Lines in $\mathbb{R}^2$ (3 parameters: w_1, w_2, b)
- Training error: 15% (makes 15 mistakes on 100 points)
- Test error: 17% (measured on fresh 1000 points)

Model B: Complex (10th-Degree Polynomial)

- Hypothesis class: Degree-10 polynomials (66 parameters!)
- Training error: 0% (perfect! Not a single mistake)
- Test error: 35% (disaster on fresh data)

⚠ Crisis

The Overfitting Catastrophe

Expected: Perfect training accuracy (100%) means excellent test performance

Reality:

- Model B training accuracy: **100%** ✓
- Model B test accuracy: **65%** ✗
- **Performance gap: 35 percentage points!**

What went wrong: Model B didn't learn the pattern—it **memorized** the training data. It fits noise instead of signal.

The fundamental problem: Training error $\hat{R}_S(h)$ is a **biased estimate** of true error $R(h)$ when we choose h based on minimizing $\hat{R}_S$.

Why does this happen?

When we have a complex hypothesis class (large $|\mathcal{H}|$), we can always find some h that fits the training data well—even if the data contains noise! The more complex $\mathcal{H}$, the more likely we find a hypothesis that “cheats” by memorizing rather than generalizing.

The Generalization Gap

Let's formalize the problem. We care about two different error rates:

📊 Comparison

Training Error (Empirical Risk):

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}[h(x_i) \neq y_i] \quad (2)$$

What we can compute—error on our sample S .

Test Error (True Risk):

$$R(h) = \text{Prob}_{(x,y) \sim \mathcal{D}}[h(x) \neq y] \quad (3)$$

What we actually care about—error on the true distribution $\mathcal{D}$.

The Generalization Gap:

$$\text{Gap} = |R(h) - \hat{R}_S(h)| \quad (4)$$

For Model B: $\text{Gap} = |0.35 - 0.00| = 0.35$ (catastrophic!)

The core insight: We can't just minimize training error. We need to **bound the generalization gap**:

$$R(h) \leq \hat{R}_S(h) + \text{generalization gap} \quad (5)$$

But how large is this gap? And how can we control it?

The Key Insight: Probabilistic Guarantees

We can't eliminate the gap entirely (the training sample is random!), but we can **bound it with high probability**.

From Week 1: Hoeffding's inequality tells us that for a fixed hypothesis h :

$$\text{Prob} [|R(h) - \hat{R}_S(h)| > \varepsilon] \leq 2 \exp(-2m\varepsilon^2) \quad (6)$$

Setting this to δ :

$$2 \exp(-2m\varepsilon^2) = \delta \quad (7)$$

$$m = \frac{1}{2\varepsilon^2} \log \frac{2}{\delta} \quad (8)$$

With $\varepsilon = 0.1$, $\delta = 0.05$, we need $m \approx 185$ samples to guarantee the gap is small.

But there's a catch! This bound only works if we choose h **before** seeing the data. When we use ERM (choosing h to minimize $\hat{R}_S$), we're choosing **after** seeing S —this creates selection bias!

Aha Moment

The Resolution: PAC Learning Framework

We can't guarantee perfection, but we can guarantee:

1. **Probably:** With confidence $1 - \delta$ (say, 95%)
2. **Approximately:** Error within ε of the best possible (say, $\varepsilon = 0.1$)
3. **Bounded samples:** Using $m = O\left(\frac{|\mathcal{H}| + \log(\frac{1}{\delta})}{\varepsilon^2}\right)$ samples

This is PAC learning: **Probably Approximately Correct** learning.

The key: Account for hypothesis class size $|\mathcal{H}|$ to prevent selection bias.

The PAC Learning Definition

Now we can formalize learning rigorously.

Definition: PAC Learning

A hypothesis class $\mathcal{H}$ is **PAC learnable** if there exists an algorithm $\mathcal{A}$ and a function $m_{\mathcal{H}} : (0, 1)^2 \rightarrow \mathbb{N}$ such that:

For every $\varepsilon > 0$ (error tolerance), $\delta > 0$ (confidence), and distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$:

If the algorithm receives $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ i.i.d. samples from $\mathcal{D}$, then with probability at least $1 - \delta$:

$$R(\hat{h}) \leq \min_{h \in \mathcal{H}} R(h) + \varepsilon \quad (9)$$

where $\hat{h}$ is the hypothesis output by $\mathcal{A}$.

The function $m_{\mathcal{H}}(\varepsilon, \delta)$ is called the **sample complexity**.

Unpacking the definition:

- **Probably** ($1 - \delta$ confidence): We can't guarantee success with probability 1 (the sample might be unlucky). But we can make failure very unlikely (say, $\delta = 0.05$).
- **Approximately** (ε -close to optimal): We don't require perfect learning. We just need to get within ε of the best hypothesis in $\mathcal{H}$.
- **Correct** (low error): The error $R(\hat{h})$ is small—close to the best achievable error in $\mathcal{H}$.
- **Efficient** (polynomial sample complexity): We need $m_{\mathcal{H}}(\varepsilon, \delta)$ to be polynomial in $\frac{1}{\varepsilon}$, $\frac{1}{\delta}$, and relevant parameters (like $|\mathcal{H}|$).

 Key Takeaway

PAC learning formalizes what it means for a hypothesis class to be “learnable”:

1. Error can be made arbitrarily small (given enough data)
2. High confidence can be achieved (failure is rare)
3. Sample complexity grows reasonably (polynomial, not exponential)

The framework converts the vague question “Can we learn this?” into a precise mathematical statement with provable guarantees.

Connection to Our Example

Let's revisit our overfitting example through the PAC lens.

Model A (Linear classifiers in $\mathbb{R}^2$):

- $|\mathcal{H}| = \infty$ (continuum of lines)
- But effective complexity is low (we'll see: VC dimension = 3)
- With 100 samples: generalization gap $\approx \sqrt{\frac{3}{100}} \approx 0.17$
- Training 15% + gap 17% $\approx$ test 17% ✓ (matches observed 17%!)

Model B (Degree-10 polynomials):

- Effective complexity is high (VC dimension ≈ 66)

- With 100 samples: generalization gap $\approx \sqrt{\frac{66}{100}} \approx 0.81$
- Training 0% + gap 81% $\approx$ test 81%... wait, we observed 35%?
- The bound is loose but **directionally correct**—high complexity $\rightarrow$ large gap

Next question: How do we compute sample complexity for specific hypothesis classes? This requires understanding **uniform convergence**—the topic of the next section.

Looking Ahead

We've seen that ERM can fail catastrophically due to selection bias. The solution is to ensure convergence happens **uniformly** over all $h \in \mathcal{H}$, not just for one fixed h .

This leads to the question: “Why do we need to prove convergence for ALL hypotheses when we only output one?”

Finite Hypothesis Classes: The Selection Bias Trap

🧐 Student Question

From the class: “We derived the PAC bound and it works for one hypothesis. But ERM chooses the best hypothesis from $\mathcal{H}$ based on training data. Since we only output **one** hypothesis $\hat{h}$, why do we need to prove convergence for **every** hypothesis in $\mathcal{H}$?”

This seems wasteful—if Hoeffding tells us each individual $\hat{R}_S(h) \rightarrow R(h)$, why isn’t that enough?

The Natural Approach: Pointwise Convergence

The Law of Large Numbers tells us that averages converge. For any **fixed** hypothesis h :

$$\text{Prob} [|R(h) - \hat{R}_S(h)| > \varepsilon] \leq 2 \exp(-2m\varepsilon^2) \quad (10)$$

Intuition: With enough samples, empirical risk $\hat{R}_S(h)$ approaches true risk $R(h)$ for each hypothesis.

Strategy: Since $\hat{h} = \text{argmin}_{h \in \mathcal{H}} \hat{R}_S(h)$ is just one hypothesis, apply Hoeffding to it:

With $m = \frac{1}{2\varepsilon^2} \log \frac{2}{\delta} \approx 185$ samples (for $\varepsilon = 0.1, \delta = 0.05$), we guarantee:

$$\text{Prob} [|R(\hat{h}) - \hat{R}_S(\hat{h})| > 0.1] \leq 0.05 \quad (11)$$

Conclusion: 185 samples should suffice, regardless of $|\mathcal{H}|$. Done, right?

Testing the Intuition: Pure Noise Experiment

Let’s test this reasoning with a concrete experiment.

Setup:

- Hypothesis class: $\mathcal{H} = \{h_1, h_2, \dots, h_{1000}\}$ (1000 binary classifiers)
- Distribution: $P[y = 1 | x] = 0.5$ for all x (pure noise—no learnable pattern!)
- True risk: $R(h_i) = 0.5$ for all i (every hypothesis is equally bad)
- Training sample: $m = 185$ points (as calculated above)

Question: What happens when we run ERM?

Expected outcome: Since all hypotheses are equally bad ($R = 0.5$), we should find $\hat{R}_S(\hat{h}) \approx 0.5$.

Actual outcome:

$$\hat{h} = \text{argmin}_{h \in \mathcal{H}} \hat{R}_S(h) \quad (12)$$

Let’s compute what we actually observe:

Comparison

Empirical risks on training data (random variation):

- h_{247} : $\hat{R}_S = 0.35 \leftarrow$ **Best on training!**
- h_{689} : $\hat{R}_S = 0.37$
- h_{142} : $\hat{R}_S = 0.38$
- ...
- h_{521} : $\hat{R}_S = 0.63 \leftarrow$ **Worst on training**

ERM chooses: $\hat{h} = h_{247}$ (training error 35%)

But remember the true risks:

- $R(h_{247}) = 0.5$ (it's actually random!)
- $R(h_{689}) = 0.5$
- All hypotheses have $R = 0.5$

Test performance: Deploy h_{247} on fresh data $\rightarrow$ 50% error (not 35%!)

What happened? With 1000 hypotheses and pure randomness, **at least one** hypothesis will appear good on the training set by pure luck—even though it's actually no better than random!

The Multiple Testing Problem

Let's calculate how likely this “lucky hypothesis” phenomenon is.

For a single hypothesis h_i :

$$\text{Prob}[\hat{R}_S(h_i) \leq 0.35] \approx 0.02 \quad (2\% \text{ chance by random variation}) \quad (13)$$

But with 1000 hypotheses, we're asking: “What's the probability that **at least one** achieves $\hat{R}_S \leq 0.35$?”

$$\text{Prob}[\exists i : \hat{R}_S(h_i) \leq 0.35] = 1 - \text{Prob}[\text{all} > 0.35] \approx 1 - (0.98)^{1000} \approx 0.9999... \quad (14)$$

Almost certain!

⚠ Crisis

The Selection Bias Catastrophe

Naive reasoning:

- Each hypothesis converges: $\hat{R}_S(h_i) \rightarrow R(h_i)$ ✓
- ERM picks best on training: $\hat{h} = \operatorname{argmin} \hat{R}_S(h)$ ✓
- Therefore $\hat{R}_S(\hat{h})$ is reliable... ✗ **WRONG!**

Reality with 1000 hypotheses on noise:

- True risks: $R(h_i) = 0.5$ for all i
- Best training risk: $\hat{R}_S(\hat{h}) = 0.35$ (looks great!)
- Actual test risk: $R(\hat{h}) = 0.5$ (disaster!)
- **Performance gap: 15 percentage points**

Probability this happens: ≈ 1.0 (virtually certain)

The fundamental flaw: Pointwise convergence $\hat{R}_S(h) \rightarrow R(h)$ for each fixed h doesn't protect ERM, which chooses h **based on** $\hat{R}_S$.

This is the **multiple testing problem**: when you test many hypotheses, at least one will appear significant by chance.

Why naive bound fails:

The Hoeffding bound for fixed h requires h to be chosen **before** seeing the data. But ERM chooses $\hat{h}$ **after** seeing S —specifically to minimize $\hat{R}_S$! This creates selection bias.

Analogy: It's like shooting arrows at a wall, then drawing targets around wherever they landed, and claiming you're an expert archer.

The Key Insight: Uniform Convergence

We need a stronger guarantee: convergence must hold **simultaneously** for all hypotheses in $\mathcal{H}$.

💡 Aha Moment

The Resolution: Uniform Convergence

Instead of asking “Does $\hat{R}_S(\hat{h})$ approximate $R(\hat{h})$?” (pointwise), we need:

“Do all empirical risks approximate all true risks simultaneously?”

Formally: Bound the **worst-case gap** over all $h \in \mathcal{H}$:

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \quad (15)$$

If this supremum is small, then **no matter which** h we choose (including via ERM), the gap is controlled.

This is **uniform convergence**: convergence that holds uniformly over the entire hypothesis class.

The Union Bound: Correcting for Multiple Hypotheses

How do we prove uniform convergence for finite $|\mathcal{H}|$?

Technique: Union bound over all hypotheses.

For each $h \in \mathcal{H}$, Hoeffding gives:

$$\text{Prob} [|R(h) - \hat{R}_S(h)| > \varepsilon] \leq 2 \exp(-2m\varepsilon^2) \quad (16)$$

Apply union bound over all $h \in \mathcal{H}$:

$$\text{Prob} [\exists h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| > \varepsilon] \leq \sum_{h \in \mathcal{H}} 2 \exp(-2m\varepsilon^2) = 2|\mathcal{H}| \exp(-2m\varepsilon^2) \quad (17)$$

Setting this to δ :

$$2|\mathcal{H}| \exp(-2m\varepsilon^2) = \delta \quad (18)$$

Solving for m :

$$m = \frac{1}{2\varepsilon^2} \log \frac{2|\mathcal{H}|}{\delta} = \frac{\log|\mathcal{H}| + \log(\frac{2}{\delta})}{2\varepsilon^2} \quad (19)$$

Comparison

Sample complexity comparison:

Pointwise (WRONG for ERM):

$$m = \frac{1}{2\varepsilon^2} \log \frac{2}{\delta} \approx 185 \quad (\text{for } \varepsilon = 0.1, \delta = 0.05) \quad (20)$$

Uniform (CORRECT for ERM):

$$m = \frac{1}{2\varepsilon^2} \left(\log |\mathcal{H}| + \log \frac{2}{\delta} \right) \quad (21)$$

For $|\mathcal{H}| = 1000$:

$$m \approx \frac{1}{0.02} (6.9 + 3.7) \approx 530 \text{ samples} \quad (22)$$

The cost of selection: $\frac{530}{185} \approx 2.9 \times$ more samples needed!

This extra cost is the price we pay for choosing h based on the data.

The PAC Bound for Finite Hypothesis Classes

We can now state the complete PAC guarantee.

Theorem: Finite Hypothesis Class PAC Bound

Let $\mathcal{H}$ be a finite hypothesis class with $|\mathcal{H}| = N < \infty$.

For any $\varepsilon, \delta > 0$ and distribution $\mathcal{D}$, if we draw $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ i.i.d. samples where:

$$m_{\mathcal{H}}(\varepsilon, \delta) = \frac{\log N + \log\left(\frac{2}{\delta}\right)}{2\varepsilon^2} \quad (23)$$

Then with probability at least $1 - \delta$ over the training sample $S \sim \mathcal{D}^m$:

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \leq \varepsilon \quad (24)$$

Corollary for ERM: The hypothesis $\hat{h} = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h)$ satisfies:

$$R(\hat{h}) \leq \hat{R}_S(\hat{h}) + \varepsilon \leq \min_{h \in \mathcal{H}} \hat{R}_S(h) + \varepsilon \leq \min_{h \in \mathcal{H}} R(h) + 2\varepsilon \quad (25)$$

ERM is approximately optimal!

🎯 Key Takeaway

Key insights:

1. Sample complexity grows with $\log|\mathcal{H}|$ (not $|\mathcal{H}|$ itself—logarithmic is good!)
2. Uniform convergence makes ERM safe: the training set winner is reliably good
3. The $\log|\mathcal{H}|$ term is the “cost of selection”—price for choosing based on data

Why logarithmic growth is powerful:

- $|\mathcal{H}| = 1000 \rightarrow \log|\mathcal{H}| \approx 7$
- $|\mathcal{H}| = 10^6 \rightarrow \log|\mathcal{H}| \approx 14$
- Even large finite classes are efficiently learnable!

Revisiting the Pure Noise Experiment

Let’s verify our bound on the noise experiment.

With uniform convergence bound ($m = 530$):

- Guarantee: $\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \leq 0.1$ with 95% confidence
- Even the best training performer: $|R(\hat{h}) - \hat{R}_S(\hat{h})| \leq 0.1$
- If $\hat{R}_S(\hat{h}) = 0.35$, then $R(\hat{h}) \in [0.25, 0.45]$
- True value $R = 0.5$ is close enough (bound is conservative but directionally correct)

With naive bound ($m = 185$):

- No protection against selection bias
- Can easily see $\hat{R}_S = 0.35$ but $R = 0.5$ (15% gap!)

🧐 Looking Ahead

We’ve solved the finite case: sample complexity = $O(\log|\mathcal{H}|)$.

But what about **infinite** hypothesis classes? Linear classifiers, neural networks, decision trees all have $|\mathcal{H}| = \infty$, making $\log|\mathcal{H}| = \infty$.

Are infinite classes unlearnable? Or are we measuring complexity wrong?

The answer: **VC dimension**—a measure of effective complexity that works even when $|\mathcal{H}| = \infty$.

VC Dimension: Beyond Parameter Counting

🤔 Student Question

From the class: “We need a measure of hypothesis class complexity to control sample complexity. Can’t we just count the number of parameters? A linear classifier in $\mathbb{R}^2$ has 3 parameters (w_1, w_2, b) , so its complexity is 3. Simple!”

This seems natural—more parameters means more flexibility, which should mean higher complexity, right?

The Natural Approach: Count Parameters

The most intuitive measure of model complexity is the number of adjustable parameters.

Strategy: Define complexity of calH as the number of free parameters.

Examples that seem to work:

- Intervals $[a, b]$ on $\mathbb{R}$: 2 parameters $\rightarrow$ complexity = 2
- Linear classifiers in $\mathbb{R}^2$: $\text{sign}(w_1x_1 + w_2x_2 + b)$ has 3 parameters $\rightarrow$ complexity = 3
- Degree- d polynomials: $d + 1$ coefficients $\rightarrow$ complexity = $d + 1$

Intuition: More parameters = more freedom to fit data = higher complexity = need more samples.

With this definition, sample complexity would be:

$$m = O\left(\frac{\# \text{ parameters}}{\varepsilon^2} \log \frac{1}{\delta}\right) \quad (26)$$

For linear classifier in $\mathbb{R}^2$: $m \approx \frac{3}{\varepsilon^2} \log \frac{1}{\delta}$. Looks reasonable!

Where This Breaks: Two Counterexamples

Let’s test parameter counting on some edge cases.

Counterexample 1: High-Frequency Oscillations

Consider the hypothesis class:

$$\text{calH} = \{\text{sign}(\sin(\omega x)) : \omega \in \mathbb{R}\} \quad (27)$$

Question: What is the complexity?

Parameter counting says: 1 parameter (ω) $\rightarrow$ complexity = 1 $\rightarrow$ very simple!

Sample complexity prediction: $m \approx \frac{1}{\varepsilon^2} \log \frac{1}{\delta} \approx 50$ samples (for $\varepsilon = 0.1, \delta = 0.05$)

Let’s test: Can we learn this with 50 samples?

Take any m points $x_1 < x_2 < \dots < x_m$ on the real line and any labeling $y_1, y_2, \dots, y_m \in \{\pm 1\}$.

Claim: We can **always** find ω such that $\text{sign}(\sin(\omega x_i)) = y_i$ for all $i = 1, \dots, m$.

Construction: Choose ω large enough so that $\sin(\omega x)$ oscillates rapidly. Between any two consecutive points x_i and x_{i+1} , the function $\sin(\omega x)$ completes many cycles. We can place the zero-crossings (where $\sin(\omega x) = 0$) precisely to match any desired labeling.

Consequence: With 1 parameter, this class can fit **any** labeling of **any** finite point set!

Comparison

Parameter counting vs reality:

Parameter counting prediction:

- Parameters: 1 (ω)
- Complexity: 1 (very simple)
- Sample complexity: $m \approx 50$ samples
- Conclusion: Easy to learn!

Actual behavior:

- Can fit arbitrary labelings of 50 points ✓
- Can fit arbitrary labelings of 100 points ✓
- Can fit arbitrary labelings of 1000 points ✓
- Can fit arbitrary labelings of **any** finite set ✓
- Conclusion: **Infinite** complexity!

The gap: Parameter counting predicts “1”, reality is “ ∞ ”

Counterexample 2: High-Dimensional But Constrained

Consider linear classifiers in $\mathbb{R}^{100}$:

$$\text{calH} = \{\text{sign}(w \cdot x + b) : w \in \mathbb{R}^{100}, b \in \mathbb{R}\} \quad (28)$$

Parameter counting says: 101 parameters $\rightarrow$ complexity = 101 $\rightarrow$ needs 5,000 samples

But here’s a puzzle: Take 3 collinear points in $\mathbb{R}^{100}$ (points on a line, say $x_1 = 0$, $x_2 = (1, 0, \dots, 0)$, $x_3 = (2, 0, \dots, 0)$).

Question: Can we realize the labeling $(+1, -1, +1)$ (positive, negative, positive)?

Answer: No! Any hyperplane in $\mathbb{R}^{100}$ is a 99-dimensional subspace. If a line intersects this hyperplane, it can cross at most **once**—meaning the line is divided into at most 2 regions. We cannot create a “sandwich” pattern with 3 collinear points.

Comparison

Collinear points example:

What parameter counting suggests:

- Parameters: 101
- Expected capability: Should be very powerful with 101 degrees of freedom
- Prediction: Can separate any configuration of 3 points

Reality:

- Can separate 3 collinear points: Only 4 out of 8 labelings possible ✕
- Cannot realize: $(+1, -1, +1)$, $(+1, -1, -1)$, $(-1, +1, +1)$, $(-1, +1, -1)$
- 101 parameters cannot even handle 3 points in special positions!

The lesson: Parameter count overstates complexity in constrained settings

The Crisis: Parameters $\neq$ Expressive Power

Crisis

The Parameter Counting Catastrophe

Parameter counting completely fails to predict learning capacity:

Failure 1: Underestimation

- $\sin(\omega x)$: 1 parameter but infinite capacity
- Can memorize arbitrary labelings of any finite set
- Sample complexity should be ∞ , not 50!

Failure 2: Overestimation

- Linear $\mathbb{R}^{100}$: 101 parameters but cannot shatter 3 collinear points
- Sample complexity should be low for this constrained structure
- Not as complex as parameter count suggests

The fundamental flaw: Parameters measure **degrees of freedom**, not **combinatorial expressiveness**.

What we need: A measure that counts “how many behaviors are possible on finite samples”, not “how many knobs can we turn”.

Why does this matter?

Using parameter counting for sample complexity:

- For $\sin(\omega x)$: Predicts $m \approx 50 \rightarrow$ Reality: Needs ∞ samples (unlearnably)
- For linear $\mathbb{R}^{100}$ on structured data: Predicts $m \approx 5000 \rightarrow$ Reality: Could work with fewer

These aren't small errors—they're **qualitatively wrong predictions** that render the theory useless.

The Key Insight: Shattering Capacity

We need to count behaviors on finite samples, not parameters.

Aha Moment

The Resolution: VC Dimension

Instead of asking “How many parameters?”, ask:

“What is the **largest set of points** that calH can label in **all possible ways**?”

This is the **shattering capacity**:

- Measures combinatorial expressiveness on finite samples
- Captures actual learning difficulty
- Works even when $|\text{calH}| = \infty$

Intuition: If calH can realize all 2^d labelings of some d points (but not $d + 1$ points), then its “effective complexity” is d —this is what determines sample complexity, not the parameter count.

The Formal Definition

Definition: VC Dimension

The **Vapnik-Chervonenkis (VC) dimension** of hypothesis class calH is:

$$\text{VCdim}(\text{calH}) = \max\{m : \text{growth}(\text{calH}, m) = 2^m\} \quad (29)$$

Equivalently: the size of the largest set that can be **shattered** by calH .

In words: VC dimension is the largest number d such that there exists at least one set of d points that calH can label in all 2^d possible ways.

For all sets of size $d + 1$, at least one labeling is impossible.

Critical distinction—common source of confusion:

- If $\text{VCdim}(\text{calH}) = d$, then there **exists** at least one set of size d that is shattered
- It is **not** required that all sets of size d be shattered (most may not be!)
- **No** set of size $d + 1$ can be shattered (this is a universal statement)

Key properties:

1. **Existence (lower bound):** At least one configuration of d points is shatterable
2. **Universality (upper bound):** No configuration of $d + 1$ points is shatterable
3. **Combinatorial:** Based on behaviors, not parameters

Revisiting the Counterexamples

Let’s apply VC dimension to our problematic cases.

Comparison

Parameter counting vs VC dimension:

Model	Parameters	VC Dimension	Match?
Intervals $[a, b]$	2	2	✓ Yes
Linear $\mathbb{R}^2$	3	3	✓ Yes
Linear $\mathbb{R}^{100}$	101	101	✓ Yes*
$\sin(\omega x)$	1	∞	✗ No!

* Match holds for general position points, but 101 parameters can't shatter 3 collinear points—showing VC dimension correctly accounts for geometric constraints.

Key insight: VC dimension correctly predicts:

- $\sin(\omega x)$ is unlearnably ($VC = \infty$)
- Linear classifiers are learnable ($VC = d + 1$ finite)
- Parameter counting fails on both!

Computing VC Dimension: Strategy

To prove $VCdim(\text{calH}) = d$, you must establish two bounds:

1. **Lower bound** ($VCdim \geq d$): Exhibit one specific set of size d that is shattered
 - Constructive: show an example configuration
 - Only need to find **one** set that works
2. **Upper bound** ($VCdim < d + 1$): Prove no set of size $d + 1$ can be shattered
 - General: prove impossibility for **all** possible sets
 - Often use structural arguments (connectivity, dimension, extremal points)

Key Takeaway

The asymmetry:

- Lower bound: Constructive (find one example that works)
- Upper bound: Universal (prove all examples fail)

This is why computing VC dimension requires both geometric intuition (for construction) and mathematical proof (for impossibility).

Example 1: Intervals on $\mathbb{R}$

Example: Intervals on the Real Line

Let $\text{calH} = \{[a, b] : a \leq b, a, b \in \mathbb{R}\}$ be axis-aligned intervals.

We predict positive inside the interval, negative outside.

Lower bound ($\text{VCdim} \geq 2$): Take two points $x_1 < x_2$.

Can we realize all 4 labelings?

- (0, 0): Use interval $[x_2 + 1, x_2 + 2]$ (both outside) ✓
- (1, 0): Use interval $[x_1 - 0.1, x_1 + 0.1]$ (only x_1 inside) ✓
- (0, 1): Use interval $[x_2 - 0.1, x_2 + 0.1]$ (only x_2 inside) ✓
- (1, 1): Use interval $[x_1 - 0.1, x_2 + 0.1]$ (both inside) ✓

All 4 dichotomies realized! Therefore $\text{VCdim} \geq 2$.

Upper bound ($\text{VCdim} < 3$): Take any three points $x_1 < x_2 < x_3$.

Can we realize the labeling $(+1, -1, +1)$ (endpoints positive, middle negative)?

Answer: No! An interval is **connected**. If x_1 and x_3 are in the interval, then every point between them (including x_2) must also be in it.

Therefore, at least one labeling (out of $2^3 = 8$) is impossible for **any** configuration of 3 points.

Conclusion: $\text{VCdim}(\text{intervals}) = 2$

Why the upper bound works: The **connectivity** property of intervals creates a structural constraint that prevents certain labelings regardless of point positions.

Example 2: Linear Classifiers in $\mathbb{R}^2$

Example: Lines in the Plane

Let $\text{calH} = \{\text{sign}(w_1x_1 + w_2x_2 + b) : w \in \mathbb{R}^2, b \in \mathbb{R}\}$ be linear classifiers in $\mathbb{R}^2$.

Lower bound ($\text{VCdim} \geq 3$): Take three points forming a triangle (non-collinear).

For any labeling of the 3 vertices (8 possibilities), we can draw a line separating the positive points from the negative points.

Examples:

- $(+, +, +)$: Any line missing all 3 points ✓
- $(+, -, -)$: Line separating first point from other two ✓
- $(+, -, +)$: Line separating middle point from endpoints ✓
- All 8 cases are realizable ✓

Figure 1: Three non-collinear points can be shattered by lines

Therefore $\text{VCdim} \geq 3$.

Upper bound ($\text{VCdim} < 4$): For any 4 points in $\mathbb{R}^2$, we cannot shatter them.

Key tool: Radon's theorem states that any $d + 2$ points in $\mathbb{R}^d$ can be partitioned into two sets whose **convex hulls** intersect.

For $\mathbb{R}^2$: Any 4 points can be partitioned into X_1, X_2 such that $\text{conv}(X_1) \cap \text{conv}(X_2) \neq \emptyset$.

If the convex hulls intersect, no line can separate X_1 from X_2 —so we cannot realize the labeling “all of X_1 positive, all of X_2 negative”.

Conclusion: $\text{VCdim}(\text{linear classifiers in } \mathbb{R}^2) = 3$

General result: For linear classifiers in $\mathbb{R}^d$:

$$\text{VCdim}(\text{hyperplanes in } \mathbb{R}^d) = d + 1 \quad (30)$$

This **happens** to match the parameter count (d dimensions + 1 bias), but this is not always the case!

Example 3: Axis-Aligned Rectangles in $\mathbb{R}^2$

Example: Rectangles in the Plane

Let $\text{calH} = \{[a_1, b_1] \times [a_2, b_2] : a_i \leq b_i\}$ be axis-aligned rectangles.

Lower bound ($\text{VCdim} \geq 4$): Take 4 points in a diamond configuration:

- $p_1 = (0, 1)$ (top)
- $p_2 = (1, 0)$ (right)
- $p_3 = (0, -1)$ (bottom)
- $p_4 = (-1, 0)$ (left)

For any labeling of these 4 points, we can construct an axis-aligned rectangle to realize it:

- To include p_1 : extend rectangle upward
- To include p_2 : extend rectangle rightward
- To include p_3 : extend rectangle downward
- To include p_4 : extend rectangle leftward

All $2^4 = 16$ dichotomies are realizable! Therefore $\text{VCdim} \geq 4$.

Upper bound ($\text{VCdim} < 5$): Take any 5 points in $\mathbb{R}^2$.

Identify extremal points:

- Leftmost point: p_{left}
- Rightmost point: p_{right}
- Topmost point: p_{top}
- Bottommost point: p_{bottom}

These 4 extreme points determine the smallest axis-aligned rectangle containing all 5 points.

Since there are 5 points but only 4 extremal positions, at least one point (call it p_{interior}) is not extremal.

Labeling that fails: Label the 4 extremal points positive (+) and p_{interior} negative (−).

Any rectangle containing all 4 extremal points must contain the entire bounding box, hence must also contain p_{interior} —contradiction!

Conclusion: $\text{VCdim}(\text{rectangles}) = 4$

Why this works: The **extremal point** argument shows that rectangles have limited complexity despite having 4 parameters.

Why VC Dimension Works

Key Takeaway

VC dimension correctly captures learning complexity:

1. **Geometric, not parametric:** Measures what configurations can be separated, not how many knobs we can turn
2. **Handles infinite classes:** Works for $|\text{calH}| = \infty$ (like linear classifiers) by counting behaviors on finite samples
3. **Accounts for constraints:** Recognizes that 101 parameters don't help if geometry constrains you (collinear points)
4. **Predicts sample complexity:** The Fundamental Theorem (next section) shows:

$$m = O\left(\frac{\text{VCdim}(\text{calH})}{\varepsilon^2} \log \frac{1}{\delta}\right) \quad (31)$$

5. **Matches intuition when it should:** For “nice” classes like hyperplanes, VC dim $\approx$ parameters, but doesn't fail when they diverge

Comparison

Sample complexity comparison:

Using parameter counting (WRONG):

$$m = O\left(\frac{\# \text{ parameters}}{\varepsilon^2} \log \frac{1}{\delta}\right) \quad (32)$$

Using VC dimension (CORRECT):

$$m = O\left(\frac{\text{VCdim}(\text{calH})}{\varepsilon^2} \log \frac{1}{\delta}\right) \quad (33)$$

For $\sin(\omega x)$:

- Parameter counting: $m \approx 50$ (predicts “learnable”)
- VC dimension: $m = \infty$ (correctly identifies “unlearnably”)

For linear $\mathbb{R}^2$:

- Parameter counting: $m \approx 150$ (using 3 parameters)
- VC dimension: $m \approx 150$ (using VC dim = 3)
- Both agree—VC dimension reduces to parameter count when appropriate!

Connection to Growth Functions

Recall from the previous section that the growth function counts:

$$\text{growth}(\text{calH}, m) = \max_{x_1, \dots, x_m} |\{(h(x_1), \dots, h(x_m)) : h \in \text{calH}\}| \quad (34)$$

VC dimension is defined as:

$$\text{VCdim}(\text{calH}) = \max\{m : \text{growth}(\text{calH}, m) = 2^m\} \quad (35)$$

Interpretation: VC dimension is the largest m where growth is maximal (2^m). After this point, growth becomes sub-exponential (polynomial!), which is what makes learning tractable.

This connection will be formalized by Sauer's lemma in the next section.

Looking Ahead

We now know VC dimension correctly measures complexity. But how does finite VC dimension lead to efficient learning?

The key: **Sauer's lemma** shows that finite VC dimension implies **polynomial growth** of the growth function:

$$\text{VCdim}(\text{calH}) = d < \infty \Rightarrow \text{growth}(\text{calH}, m) \leq \left(\frac{em}{d}\right)^d = O(m^d) \quad (36)$$

This polynomial bound (instead of exponential 2^m) is what makes learning with infinite hypothesis classes tractable.

Sauer's Lemma: From Exponential to Polynomial

We now know VC dimension correctly measures complexity. But a crucial question remains: **why does finite VC dimension guarantee learnability?**

The answer: Sauer's lemma proves that finite VC dimension transforms exponential growth into polynomial growth—the mathematical bridge from geometry to learning theory.

The Direct Approach: Use Growth Functions

Recall our PAC bound for finite hypothesis classes:

$$m = O\left(\frac{\log|\text{calH}| + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (37)$$

For infinite classes ($|\text{calH}| = \infty$), we can't use $\log|\text{calH}|$ directly. But we developed **growth functions** to count behaviors:

$$\text{growth}(\text{calH}, m) = \max_{x_1, \dots, x_m} |\{(h(x_1), \dots, h(x_m)) : h \in \text{calH}\}| \quad (38)$$

Natural idea: Replace $\log|\text{calH}|$ with $\log \text{growth}(\text{calH}, m)$ in the bound.

Strategy: Use union bound over $\text{growth}(\text{calH}, m)$ behaviors instead of $|\text{calH}|$ hypotheses:

$$m = O\left(\frac{\log \text{growth}(\text{calH}, m) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (39)$$

This seems like a solution! We've replaced counting hypotheses with counting behaviors.

The Structural Obstacle: Exponential Growth

⚠ Crisis

The Exponential Growth Trap

The fatal flaw: For most infinite hypothesis classes, $\text{growth}(\text{calH}, m) = 2^m$ (exponential)!

What this means:

- Linear classifiers in $\mathbb{R}^2$: Can realize all $2^3 = 8$ labelings of 3 points
- Neural networks: Can realize all 2^m labelings for many m values
- Even “simple” infinite classes: Often have exponential growth

Sample complexity with exponential growth:

$$m = O\left(\frac{\log(2^m) + \log(\frac{1}{\delta})}{\varepsilon^2}\right) = O\left(\frac{m + \log(\frac{1}{\delta})}{\varepsilon^2}\right) \quad (40)$$

This is **circular!** The sample size m appears on both sides, making the bound **vacuous**.

The impossibility: Exponential growth $\rightarrow$ need exponential samples $\rightarrow$ unlearnably!

Let’s see the disaster concretely.

Example: Linear classifiers in $\mathbb{R}^2$ (VC dim = 3).

For $m = 10$ points:

- Growth function: Could be as high as $2^{10} = 1024$
- Using $\log \text{growth}(10) = \log(1024) = 10$ in the bound:

$$m = O\left(\frac{10 + 3.7}{0.02}\right) \approx 685 \text{ samples} \quad (41)$$

But wait—we used $m = 10$ to compute growth, yet the bound says we need $m = 685$!

Try again with $m = 685$:

- Growth: Could be 2^{685} (if we can shatter this many)
- Bound: $m = O\left(\frac{685 + 3.7}{0.02}\right) \approx 34,435$ samples

Infinite regress! Each time we increase m , the growth function potentially increases exponentially, requiring even more samples.

Comparison

The exponential growth disaster:

Quantity	What It Is	Can Compute?	Tractable?
True risk $R(h)$	$\text{Exp}[\text{loss}(h, z)]$	× No (unknown $\mathcal{D}$)	—
Growth 2^m	Behaviors on m points	✓ Yes	× No (exponential)
Sample bound	$O(\frac{m}{\epsilon^2})$	—	× No (vacuous)

The fundamental obstacle: Even though growth function is computable (unlike true risk), if it grows exponentially, the resulting sample complexity is exponential—meaning learning is impossible in practice.

What we need: A way to show growth is **not** exponential when VC dimension is finite.

The Key Transformation: VC Dimension → Polynomial Growth

Here's the critical insight: finite VC dimension **prevents** exponential growth.

Intuition: If $\text{VCdim}(\text{calH}) = d$, then:

- We can shatter **some** sets of size d (giving 2^d behaviors)
- We **cannot** shatter **any** set of size $d + 1$ (limiting growth)

This “failure to shatter” constraint should force growth to slow down from exponential to polynomial.

Aha Moment

The Resolution: Sauer's Lemma

Finite VC dimension implies polynomial growth:

If $\text{VCdim}(\text{calH}) = d < \infty$, then:

$$\text{growth}(\text{calH}, m) \leq \sum_{i=0}^d \binom{m}{i} \leq \left(\frac{em}{d}\right)^d = O(m^d) \quad (42)$$

The transformation:

- Before: $\text{growth}(\text{calH}, m) = 2^m$ (exponential, unlearnably)
- After: $\text{growth}(\text{calH}, m) \leq O(m^d)$ (polynomial, tractable!)

This converts the circular bound into a meaningful guarantee.

Why This Transformation Works

Let's see the power of polynomial growth.

Example: Linear classifiers in $\mathbb{R}^2$ (VC dim = 3).

With exponential bound (before Sauer):

- $m = 10$: $\text{growth}(10)$ could be $2^{10} = 1024$
- Sample bound: $m = O\left(\frac{\log 1024}{\varepsilon^2}\right) = O\left(\frac{10}{\varepsilon^2}\right)$ (circular!)

With polynomial bound (after Sauer):

- $m = 10$: $\text{growth}(10) \leq \binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3} = 1 + 10 + 45 + 120 = 176$
- Much smaller than $2^{10} = 1024$!

Using $\log(176) \approx 7.5$:

$$m = O\left(\frac{7.5 + \log(\frac{1}{\delta})}{\varepsilon^2}\right) = O\left(\frac{3 \log m + \log(\frac{1}{\delta})}{\varepsilon^2}\right) \quad (43)$$

This is **non-circular**! The $\log m$ term grows slowly, so we can solve for m .

 Comparison
Sample complexity transformation:**Before Sauer's lemma:**

$$m = O\left(\frac{\log(2^m) + \log(\frac{1}{\delta})}{\varepsilon^2}\right) = O\left(\frac{m + \log(\frac{1}{\delta})}{\varepsilon^2}\right) \quad (44)$$

- Growth: Exponential (2^m)
- Bound: Vacuous (circular in m)
- Conclusion: Cannot learn

After Sauer's lemma:

$$m = O\left(\frac{\log(m^d) + \log(\frac{1}{\delta})}{\varepsilon^2}\right) = O\left(\frac{d \log m + \log(\frac{1}{\delta})}{\varepsilon^2}\right) \quad (45)$$

- Growth: Polynomial ($O(m^d)$)
- Bound: Non-vacuous (logarithmic in m)
- Conclusion: **Can learn with $m = O\left(\frac{d}{\varepsilon^2}\right)$ samples!**

The key difference: $\log(2^m) = m$ (linear) vs $\log(m^d) = d \log m$ (logarithmic in m)

Sauer's Lemma: The Formal Statement

Theorem: Sauer-Shelah Lemma

Let calH be a hypothesis class with $\text{VCdim}(\text{calH}) = d < \infty$.

Then for all $m \in \mathbb{N}$:

$$\text{growth}(\text{calH}, m) \leq \sum_{i=0}^d \binom{m}{i} \quad (46)$$

For $m \geq d$, this sum is at most:

$$\left(\frac{em}{d}\right)^d = O(m^d) \quad (47)$$

where $e \approx 2.718$ is Euler's number.

Unpacking the bound:

1. **Exact form:** $\sum_{i=0}^d \binom{m}{i}$ sums the first $d + 1$ binomial coefficients
 - Counts subsets of size at most d
 - Always less than 2^m when $d < m$
2. **Asymptotic form:** $\left(e\frac{m}{d}\right)^d$ is a polynomial of degree d in m
 - Easy to work with in sample complexity analysis
 - Shows growth rate clearly: degree d polynomial
3. **Two regimes:**
 - If $\text{VCdim}(\text{calH}) = d < \infty$: $\text{growth}(\text{calH}, m) = O(m^d)$ (polynomial)
 - If $\text{VCdim}(\text{calH}) = \infty$: $\text{growth}(\text{calH}, m) = 2^m$ for infinitely many m (exponential)

Numerical Verification

Let's verify Sauer's bound on our examples.

Comparison

Linear classifiers in $\mathbb{R}^2$ (VC dim = 3):

m	2^m (exponential)	$\sum_{i=0}^3 \binom{m}{i}$ (Sauer exact)	$(e^{\frac{m}{3}})^3$ (Sauer asymptotic)	Ratio
3	8	8	7.4	1.0
5	32	26	25.4	0.81
10	1024	176	246	0.17
20	1,048,576	1,351	1,974	0.0013
100	$\approx 10^{30}$	166,751	166,375	$\approx 10^{-25}$

Key observations:

1. Sauer's bound is **tight** at $m = d$ (both give $2^3 = 8$)
2. As m increases, exponential 2^m explodes while polynomial $O(m^3)$ grows slowly
3. At $m = 100$: exponential is 10^{30} but polynomial is only $\approx 166,000$
4. The gap is the difference between unlearnably and tractable!

The Impact on Sample Complexity

Now we can complete the PAC learning story for infinite classes.

Sample complexity with VC dimension:

Starting from the agnostic finite bound:

$$m = O\left(\frac{\log|\text{calH}| + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (48)$$

Replace $\log|\text{calH}|$ with $\log \text{growth}(\text{calH}, 2m)$ (using symmetrization—see Week 3):

$$m = O\left(\frac{\log \text{growth}(\text{calH}, 2m) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (49)$$

Apply Sauer's lemma with $d = \text{VCdim}(\text{calH})$:

$$m = O\left(\frac{\log((2m)^d) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) = O\left(\frac{d \log m + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (50)$$

Solving for m (the $\log m$ term is absorbed):

$$m = O\left(\frac{d + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (51)$$

Key Takeaway

The final PAC bound for VC dimension:

To learn hypothesis class calH with $\text{VCdim}(\text{calH}) = d < \infty$:

$$m = \Theta\left(\frac{d + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (52)$$

What this means:

1. Sample complexity depends on VC dimension d , not $|\text{calH}|$
2. Infinite classes ($|\text{calH}| = \infty$) are learnable if $d < \infty$
3. Growth is linear in d and $\frac{1}{\varepsilon^2}$ (very efficient!)
4. Logarithmic in $\frac{1}{\delta}$ (high confidence is cheap)

The transformation is complete: Geometry (VC dimension) $\rightarrow$ Combinatorics (polynomial growth) $\rightarrow$ Learning (sample complexity)

Proof Sketch of Sauer's Lemma

Proof strategy: Induction on $m + d$ with a clever projection argument.

Base cases:

- If $m = 1$: Any class can realize at most 2 labelings of 1 point, so $\text{growth}(\text{calH}, 1) \leq 2 = \binom{1}{0} + \binom{1}{1} \checkmark$
- If $d = 0$: $\text{VCdim} = 0$ means cannot shatter any point, so $\text{growth}(\text{calH}, m) \leq 1 = \binom{m}{0} \checkmark$

Inductive step: Assume bound holds for $(m-1, d)$ and $(m-1, d-1)$. We'll prove it for (m, d) .

Take any set $S = \{x_1, \dots, x_m\}$ and consider $S' = \{x_1, \dots, x_{m-1}\}$ (remove last point).

For each dichotomy on S , project it to S' by ignoring x_m 's label. Two cases:

1. **Unique projection:** The labeling of S' extends to only one labeling of S
 - These contribute at most $\text{growth}(\text{calH}, m-1)$ dichotomies
2. **Duplicate projection:** The labeling of S' extends to **both** labelings of S (differs only on x_m)
 - Let $\mathcal{B}$ be the set of such "shattered extensions"
 - Key insight: If $\mathcal{B}$ could shatter a set of size d in S' , then adding x_m would give a shattered set of size $d+1$ in S —contradiction since $\text{VCdim}(\text{calH}) = d$!
 - Therefore $\text{VCdim}(\mathcal{B}) \leq d-1$
 - By induction: $|\mathcal{B}| \leq \text{growth}(\text{calH}, m-1, d-1)$

Combining both cases:

$$\text{growth}(\text{calH}, m) \leq \text{growth}(\text{calH}, m-1) + \text{growth}(\text{calH}, m-1, d-1) \quad (53)$$

This recurrence is exactly Pascal's identity for binomial coefficients!

$$\binom{m}{i} = \binom{m-1}{i} + \binom{m-1}{i-1} \quad (54)$$

By induction:

$$\text{growth}(\text{calH}, m) \leq \sum_{i=0}^d \binom{m}{i} \quad (55)$$

The asymptotic bound $(e \frac{m}{d})^d$ follows from standard binomial estimates. qed

Connection

Proof technique: This is a beautiful example of:

1. **Double induction** on $m + d$
2. **Projection argument** removing one point and analyzing structure
3. **Combinatorial identity** connecting recurrence to binomials

The elegance: VC dimension's “cannot shatter $d + 1$ ” property translates directly into the recurrence relation that yields polynomial growth.

Why Polynomial vs Exponential Matters

The difference between $O(m^d)$ and $O(2^m)$ is the difference between tractable and intractable.

Comparison

Credit scoring example (VC dim = 11):

With exponential growth (2^m):

- $m = 1000$ samples: growth $\approx 2^{1000} \approx 10^{301}$ behaviors
- Sample bound: $m = O\left(\frac{\log(10^{301})}{\epsilon^2}\right) = O\left(\frac{301}{0.02}\right) \approx 15,000$ samples
- Try $m = 15,000$: growth $\approx 2^{15,000}$ (impossibly large!)
- Conclusion: Cannot learn

With polynomial growth ($O(m^{11})$):

- $m = 1000$ samples: growth $\leq \left(\frac{e \cdot 1000}{11}\right)^{11} \approx 10^{24}$ behaviors
- Sample bound: $m = O\left(\frac{\log(10^{24})}{\epsilon^2}\right) = O\left(\frac{24 \cdot 2.3}{0.02}\right) \approx 2,760$ samples
- This is stable! With $m = 2,760$: growth $\approx 10^{26}$ (grows slowly)
- Conclusion: **Can learn with 3,000 samples!**

The transformation: 10^{301} (astronomical) $\rightarrow 10^{24}$ (large but manageable)

This 10^{277} gap is what Sauer's lemma provides.

Connection to the Fundamental Theorem

Sauer's lemma is the key step in proving the Fundamental Theorem of PAC Learning.

Theorem: Fundamental Theorem of PAC Learning (Informal)

A hypothesis class $\mathcal{H}$ is PAC-learnable if and only if $\text{VCdim}(\mathcal{H}) < \infty$.

Moreover, sample complexity is:

$$m = \Theta\left(\frac{\text{VCdim}(\mathcal{H}) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (56)$$

Proof roadmap:

1. **Forward direction** (finite VC $\rightarrow$ learnable):
 - Sauer's lemma: $\text{VCdim} < \infty \Rightarrow \text{growth}(m) = O(m^d)$ ✓ (this section)
 - Uniform convergence: Polynomial growth $\rightarrow$ finite sample complexity
 - Result: $m = O\left(\frac{d}{\varepsilon^2}\right)$ samples suffice
2. **Reverse direction** (infinite VC $\rightarrow$ unlearnably):
 - $\text{VCdim} = \infty \Rightarrow$ can shatter arbitrarily large sets
 - No algorithm can learn with finite samples (no free lunch)
 - Result: Learning impossible

Sauer's lemma establishes the forward direction—the constructive, positive result that finite VC dimension enables learning.

 Looking Ahead

We've now completed the core theory:

1. PAC framework formalizes learning
2. Uniform convergence prevents selection bias
3. VC dimension measures true complexity
4. Sauer's lemma transforms exponential $\rightarrow$ polynomial

Next step: Week 3 develops **Rademacher complexity** for data-dependent bounds, proves the complete Fundamental Theorem (both directions), and introduces the **symmetrization technique** that makes it all rigorous.

The journey from here: tighter bounds, algorithmic considerations, and extensions beyond binary classification.

Week 2 Synthesis: The Learning Theory Pipeline

We've now built the complete PAC learning framework:

The Four Key Ideas

Concept	Key Insight	What It Prevents
PAC Framework	Probabilistic guarantees with explicit confidence and accuracy parameters	Overfitting: Training accuracy alone is misleading
Uniform Convergence	Simultaneous bounds over all hypotheses in $\mathcal{H}$	Selection bias: Choosing h based on data without accounting for multiplicity
VC Dimension	Combinatorial measure of shattering capacity	Parameter counting fallacy: Parameters $\neq$ expressiveness
Sauer's Lemma	Finite VC dimension $\rightarrow$ polynomial growth	Exponential barrier: Makes infinite classes tractable

The Complete Picture

The learning theory pipeline connects four domains:

$$\text{Geometry} \longrightarrow \text{Combinatorics} \longrightarrow \text{Statistics} \longrightarrow \text{Learning} \quad (57)$$

Concretely:

1. **Geometry:** VC dimension measures shattering capacity ($\text{VCdim}(\mathcal{H}) = d$)
2. **Combinatorics:** Sauer's lemma bounds growth ($\text{growth}(m) \leq O(m^d)$)
3. **Statistics:** Uniform convergence gives guarantees ($\sup_h |R(h) - \hat{R}(h)| \leq \varepsilon$)
4. **Learning:** Sample complexity determined ($m = O(\frac{d}{\varepsilon^2})$)

Sample Complexity Hierarchy

Setting	Complexity	Rate	Key Property
Finite, realizable	$O\left(\frac{\log \text{calH} }{\varepsilon}\right)$	$\frac{1}{\varepsilon}$	Optimal for noiseless
Finite, agnostic	$O\left(\frac{\log \text{calH} }{\varepsilon^2}\right)$	$\frac{1}{\varepsilon^2}$	Cost of noise robustness
Infinite, finite VC	$O\left(\frac{d}{\varepsilon^2}\right)$	$\frac{1}{\varepsilon^2}$	Replaces $\log \text{calH} $ with d
Infinite VC	∞	—	Not PAC-learnable

Key takeaway: VC dimension completely characterizes PAC learnability.

Worked Example: Linear Classifiers in $\mathbb{R}^{10}$

Let's apply everything we've learned to credit scoring.

Setup:

- Features: 10 variables (income, debt, credit history, etc.)
- Hypothesis class: Linear classifiers $\text{calH} = \{\text{sign}(w \cdot x + b) : w \in \mathbb{R}^{10}, b \in \mathbb{R}\}$
- Goal: Predict loan default with $\varepsilon = 0.05$ error and $\delta = 0.05$ confidence

Analysis:

1. **VC dimension:** $\text{VCdim}(\text{calH}) = 11$ (for hyperplanes in $\mathbb{R}^{10}$)
2. **Sample complexity:**

$$m = O\left(\frac{d + \log(\frac{1}{\delta})}{\varepsilon^2}\right) = O\left(\frac{11 + \log(20)}{0.0025}\right) \approx 5,600 \text{ samples} \quad (58)$$

3. **Growth function at $m = 5,600$:**

$$\text{growth}(\text{calH}, 5600) \leq \left(\frac{e \cdot 5600^{11}}{11}\right) \approx 10^{30} \quad (59)$$

Compare to exponential: $2^{5600} \approx 10^{1686}$ (astronomical!)

Sauer's lemma reduced complexity by factor of 10^{1656} —this is why learning works.

4. **Verification:**

- Train ERM on 5,600 examples
- With 95% confidence: true error within 5% of empirical error
- If $\hat{R}(\hat{h}) = 0.08$, then $R(\hat{h}) \in [0.03, 0.13]$ with high probability

What we've guaranteed: Despite having infinite hypothesis space and noisy data, we can reliably learn with 5,600 samples—far fewer than the naive exponential barrier.

Common Misconceptions Resolved

Wrong Belief	Correct Understanding
"Training accuracy = test accuracy"	Training can be misleading due to overfitting (35% gap possible!)
"One good hypothesis is enough"	Must account for selection bias (P≈1 of lucky hypothesis with 1000 choices)
"Complexity = number of parameters"	Complexity = shattering capacity ($\sin(\omega x)$: 1 param but VC = ∞)
"Infinite classes are unlearnably"	Finite VC dimension makes them learnable (VC = 11 for linear $\mathbb{R}^{10}$)
"More data always helps"	Only up to $O(\frac{d}{\varepsilon^2})$ —diminishing returns after that

Looking Ahead to Week 3

Next week we complete the story:

1. **Rademacher Complexity:** Data-dependent bounds (tighter than VC)
2. **Fundamental Theorem:** Complete characterization (both directions)
3. **Symmetrization:** The technique making rigorous proofs possible
4. **No Free Lunch:** Why infinite VC dimension prevents learning

The big question: We've shown $\text{finite VC} \rightarrow \text{learnable}$. But is the converse true? Can we prove $\text{infinite VC} \rightarrow \text{unlearnable}$?

The answer connects to deep results about the limits of learning—what's possible and what's fundamentally impossible.

Confidence Check

Before moving to Week 3, ensure you can:

- ☐ Explain the overfitting crisis with numerical examples
- ☐ Compute sample complexity using VC dimension
- ☐ Prove VC dimension for simple hypothesis classes
- ☐ State Sauer's lemma and explain why polynomial growth matters

If any checkbox is unclear, review the relevant section!